THE AGENTIC AUDIENCES BLUEPRINT

Bridging the gap between today's RTB ecosystem and tomorrow's embedding-native advertising stack.

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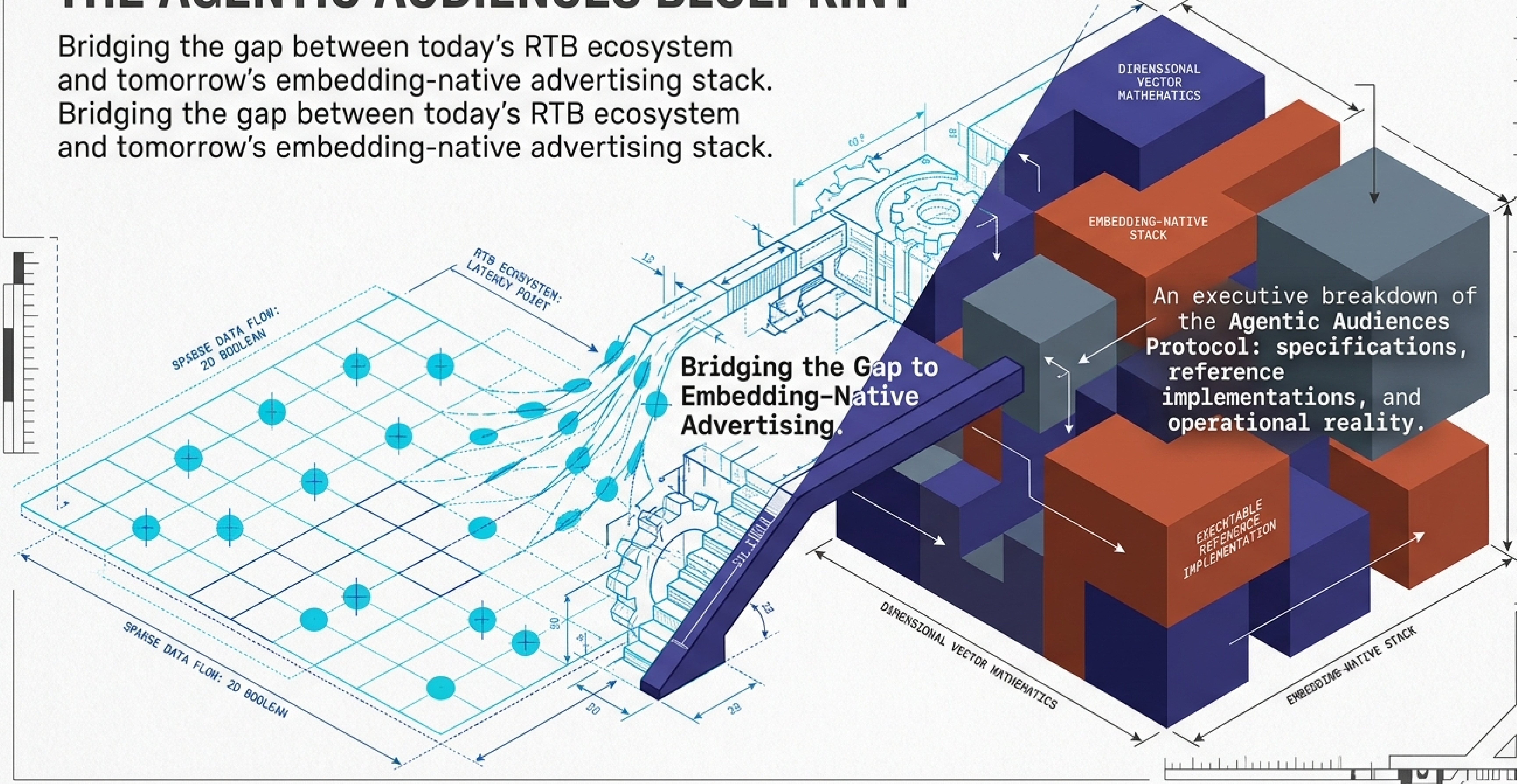
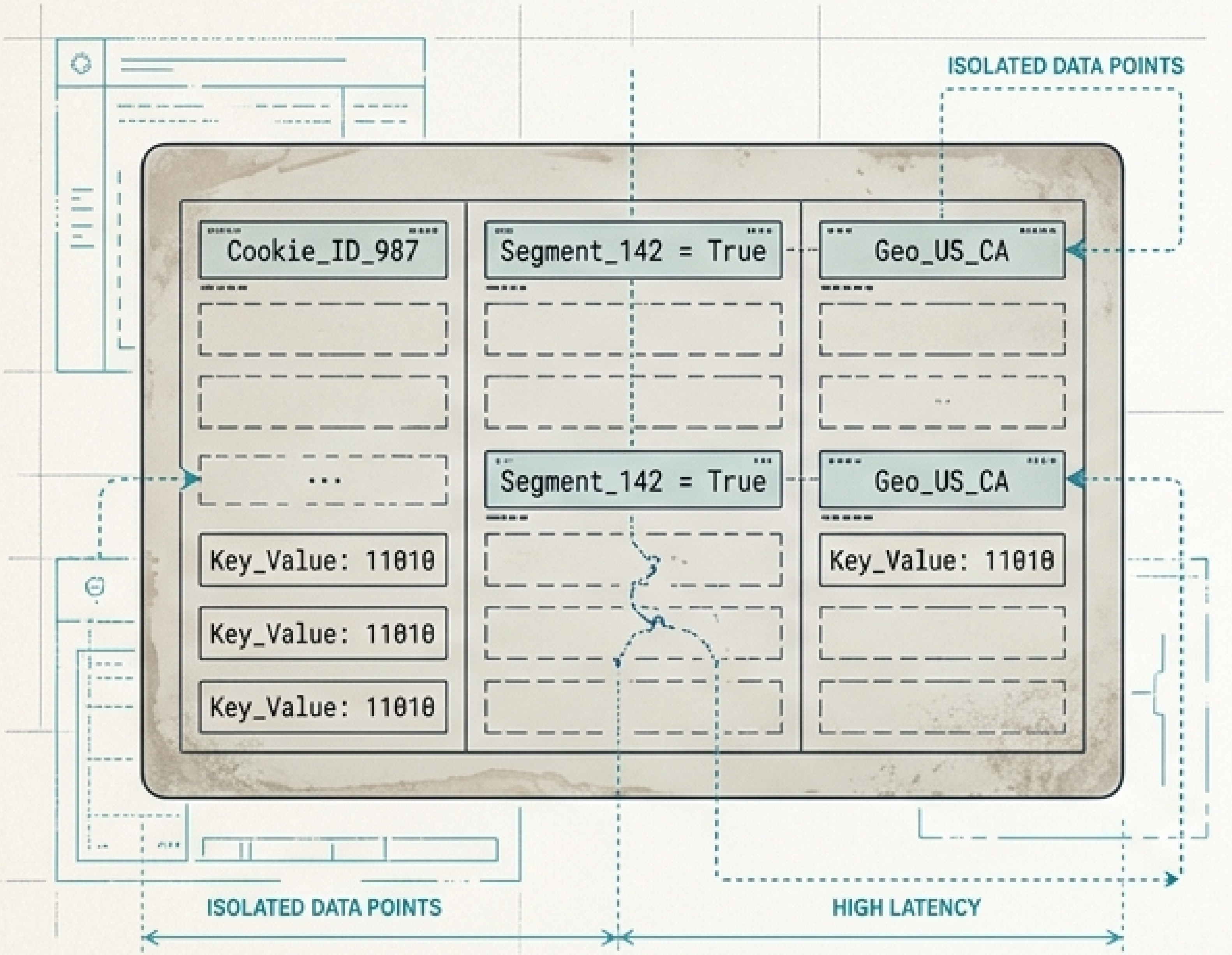


Bridging the Gap to Embedding-Native Advertising.

An executive breakdown of the **Agentic Audiences Protocol: specifications,** reference implementations, and operational reality.

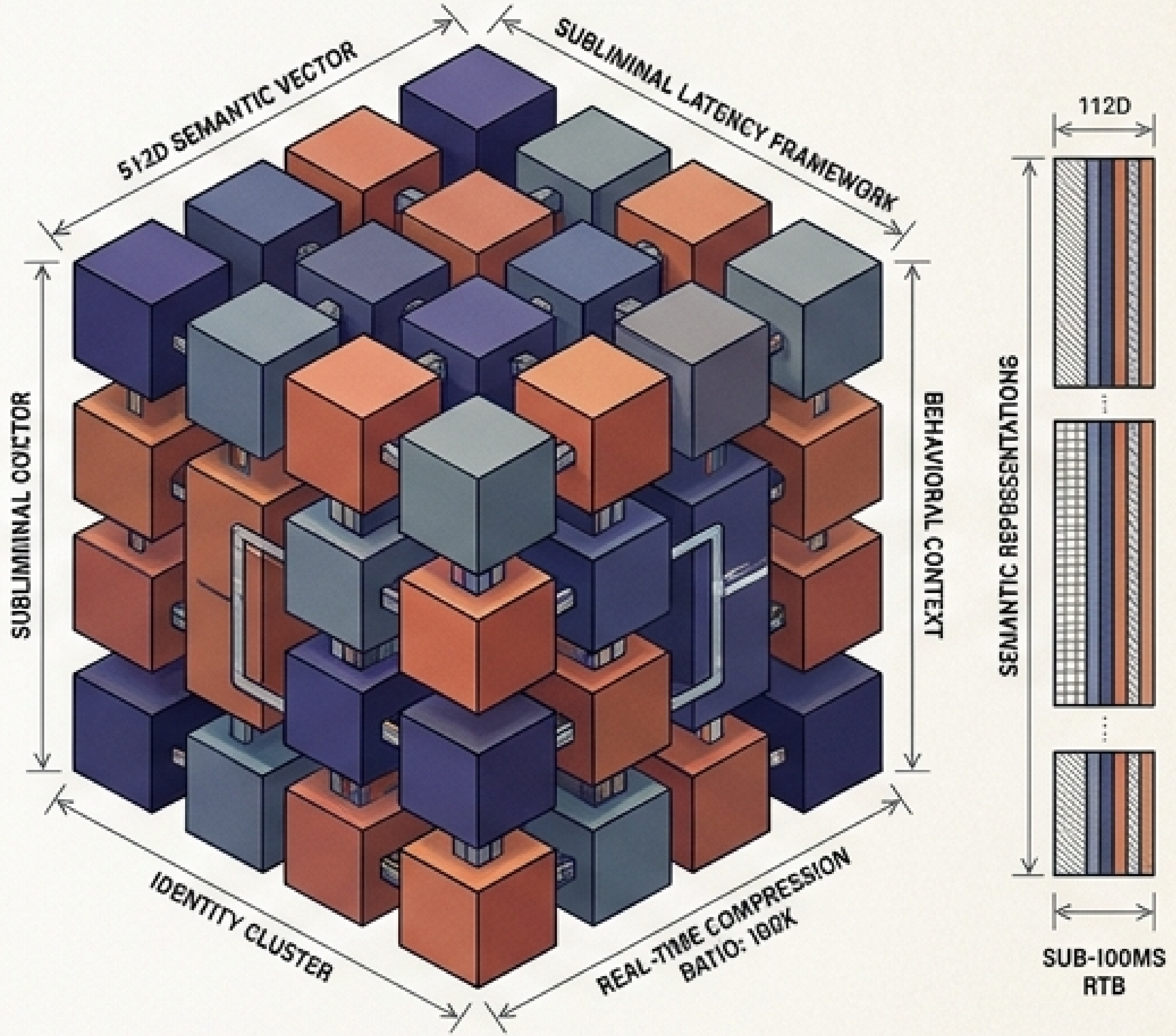
Sparse identifiers are failing as interfaces for AI-native advertising models.

THE PAST: SPARSE SIGNALS



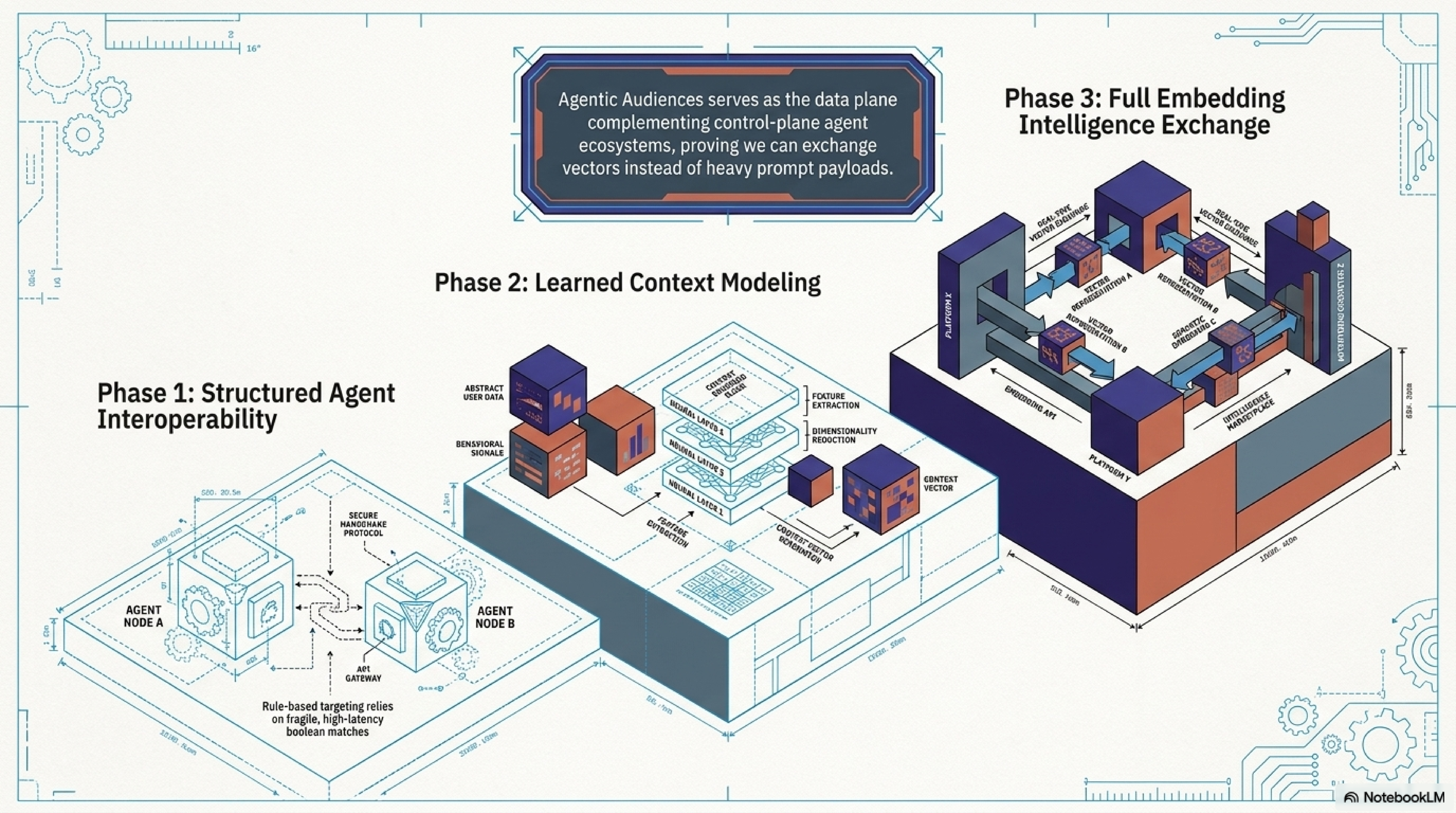
Rule-based targeting relies on fragile, high-latency boolean matches.

THE FUTURE: DENSE REPRESENTATIONS

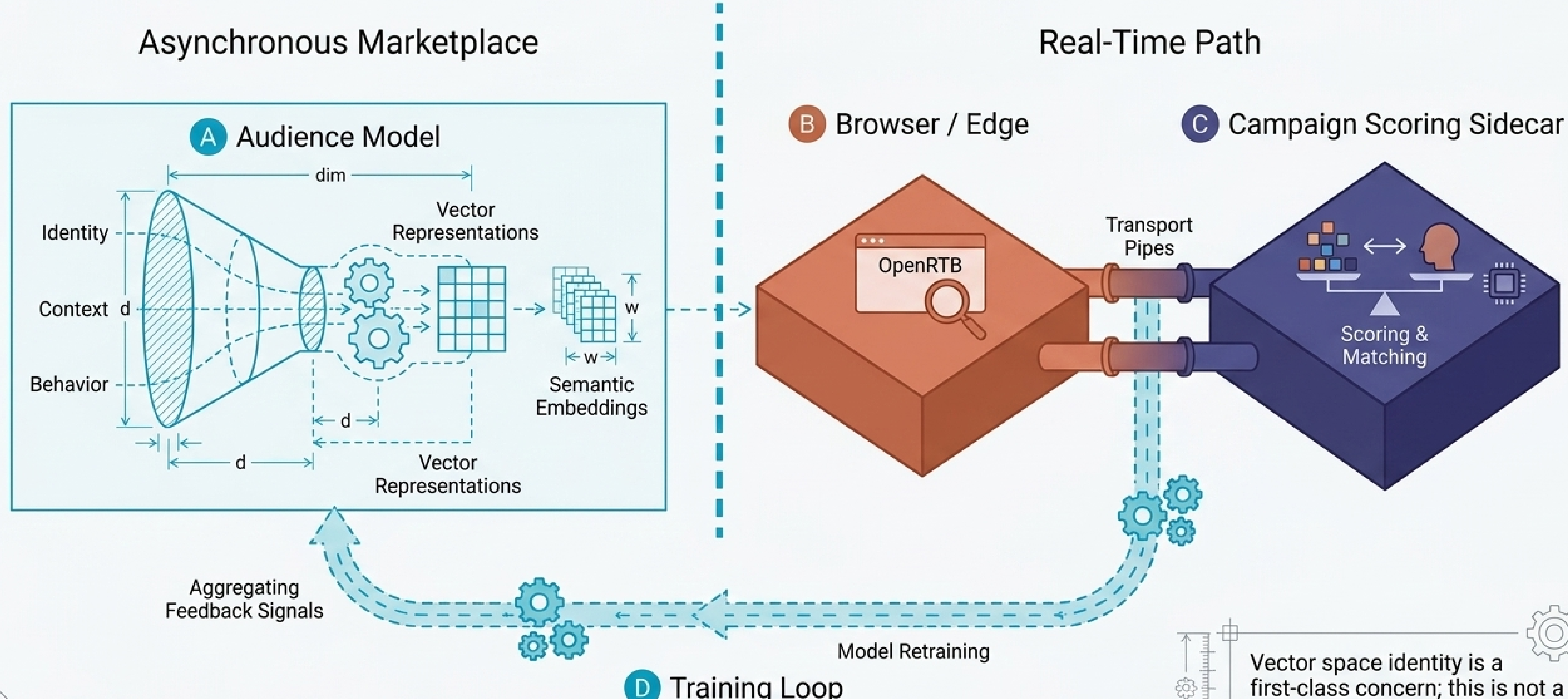


AI systems require fast similarity math over semantic representations, compressing context, identity, and behavior while adhering to sub-100ms RTB constraints.

NotebookLM


Agentic Audiences serves as the data plane complementing control-plane agent ecosystems, proving we can exchange vectors instead of heavy prompt payloads.
Phase 3: Full Embedding Intelligence Exchange
Phase 2: Learned Context Modeling
Phase 1: Structured Agent Interoperability
ABSTRACT USER DATA
FEATURE EXTRACTION
DIMENSIONALITY REDUCTION
SECURE
PROTOCOL
AGENT NODE A
AGENT NODE B
GATEWAY
Rule-based targeting relies on fragile, high-latency boolean matches
EMBEDDING API
PLATFORM Y

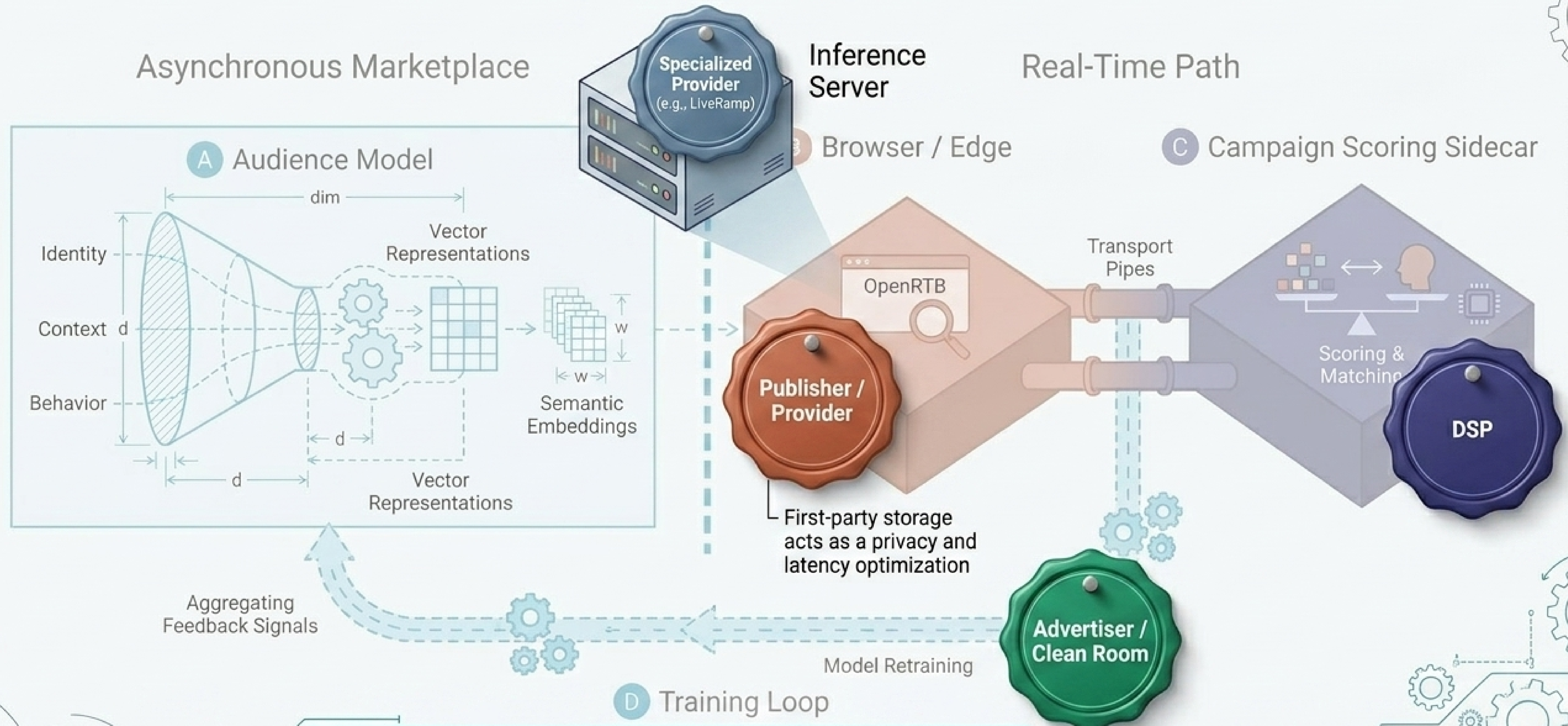
Embeddings decouple signal generation from real-time bidding constraints



Vector space identity is a first-class concern; this is not a monolithic single-vendor system

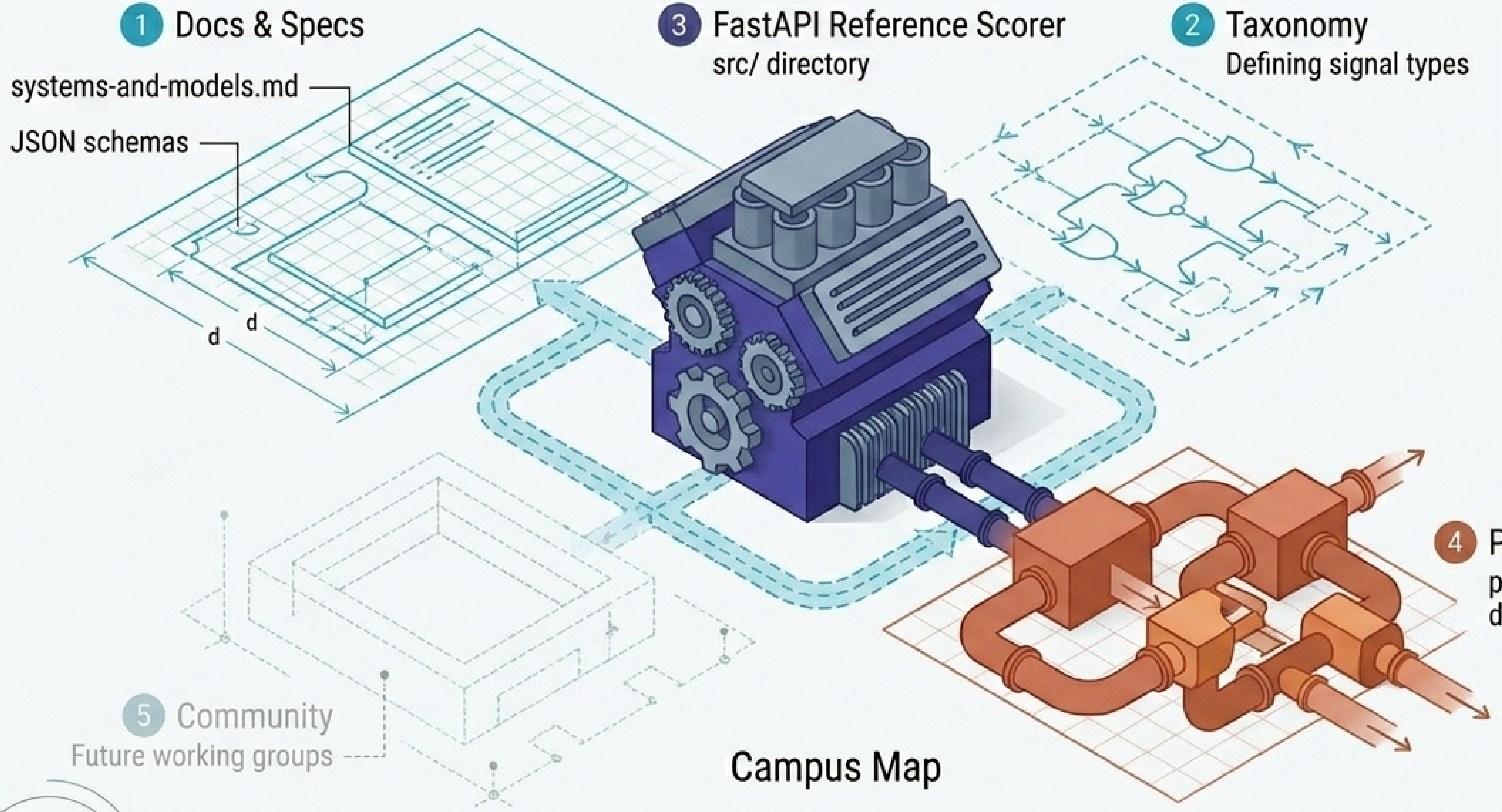
NotebookLM

The architecture requires a federated ecosystem rather than a monolithic product



Takeaway: The open protocol acts as the strict interoperability boundary between these independently operated silos.

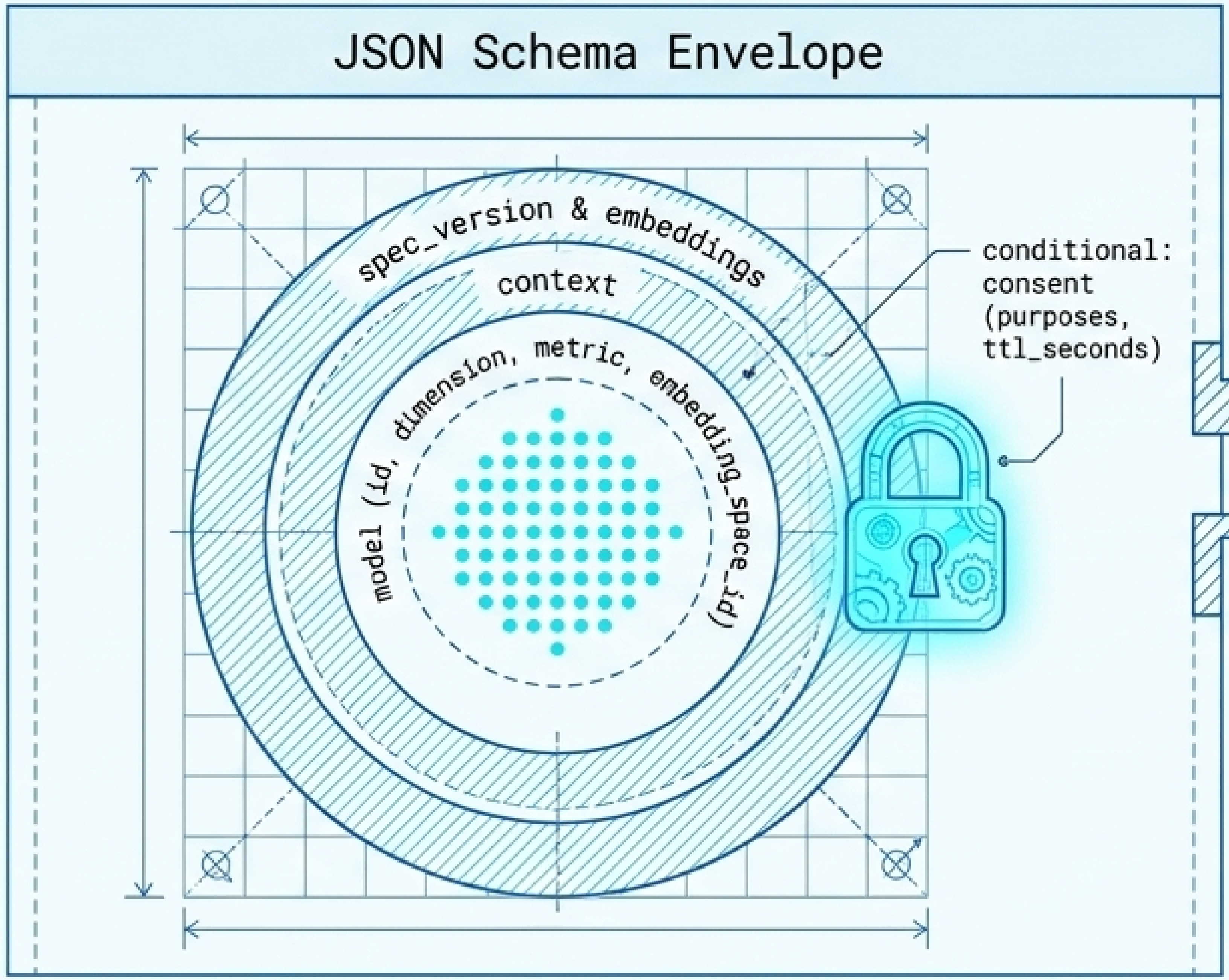
The repository operates simultaneously as a forward-looking standard and a concrete prototype.

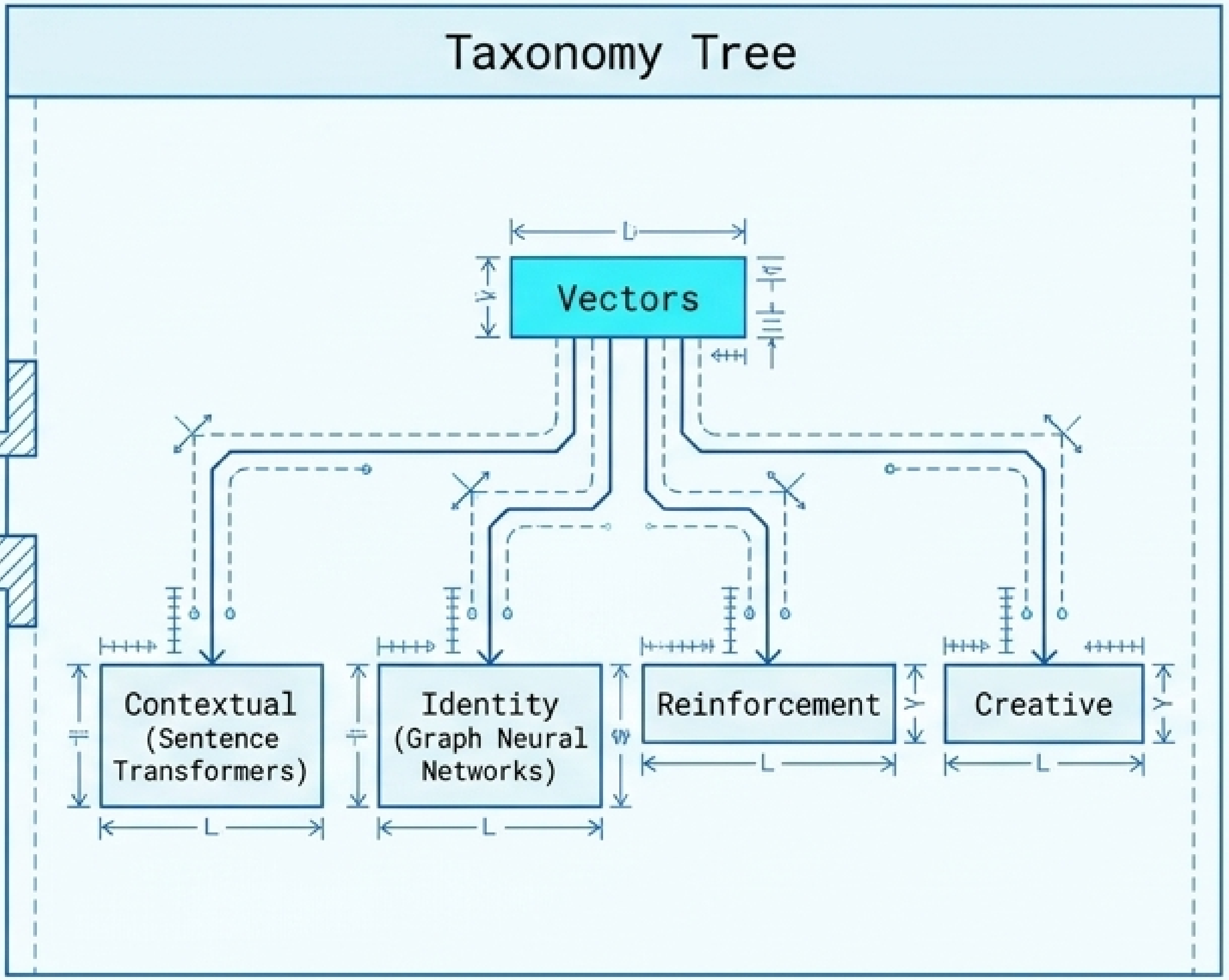


Blueprint zones represent draft standards.

Solid structures represent executable, mature code.

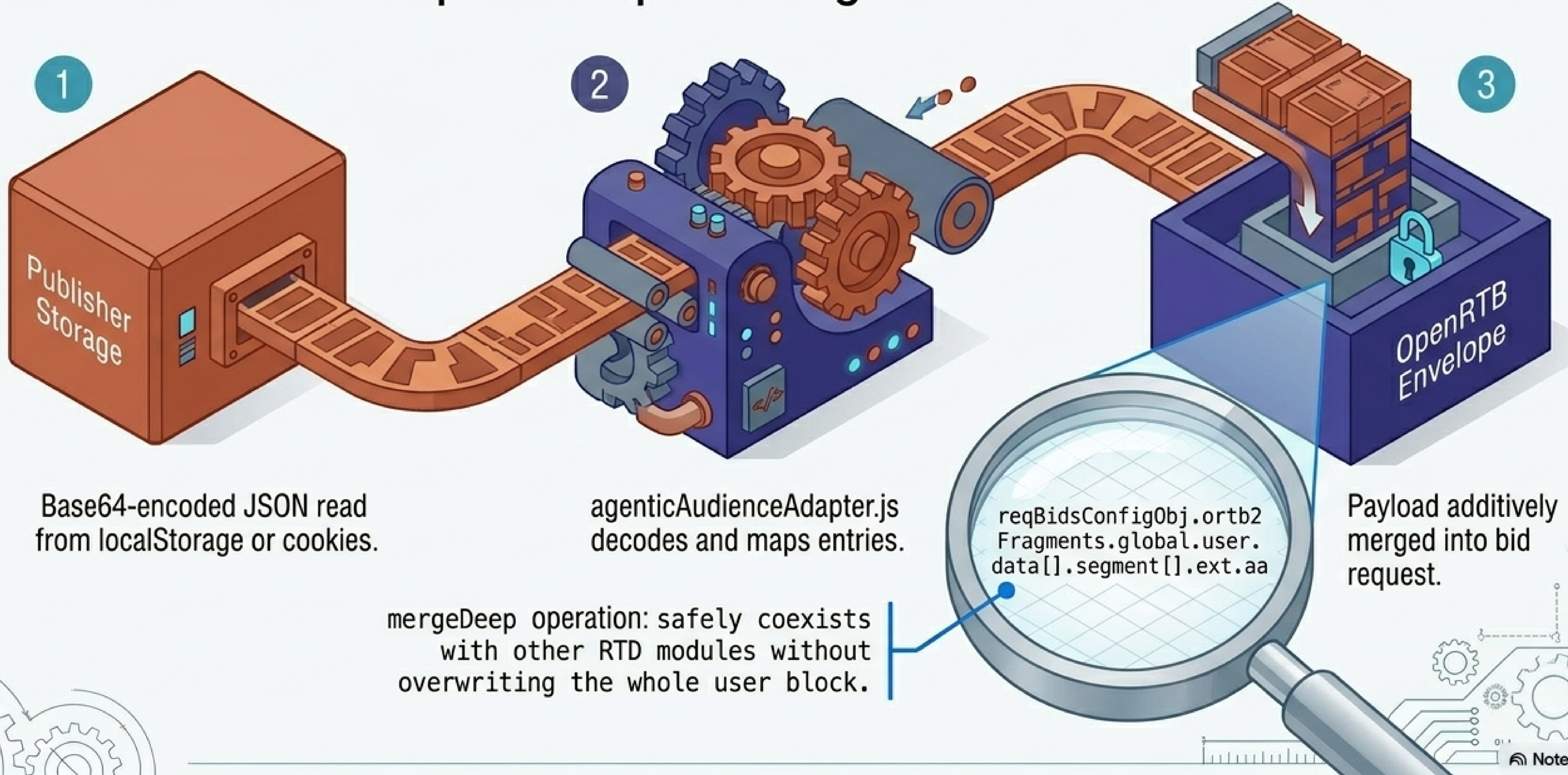
The specification layer defines strict, policy-aware envelopes for embedding exchange.





Compatibility is conditional. Embeddings carry rich metadata to judge comparability before scoring occurs.

The Prebid RTD module safely injects stored embeddings into standard OpenRTB plumbing.



Base64-encoded JSON read from localStorage or cookies.

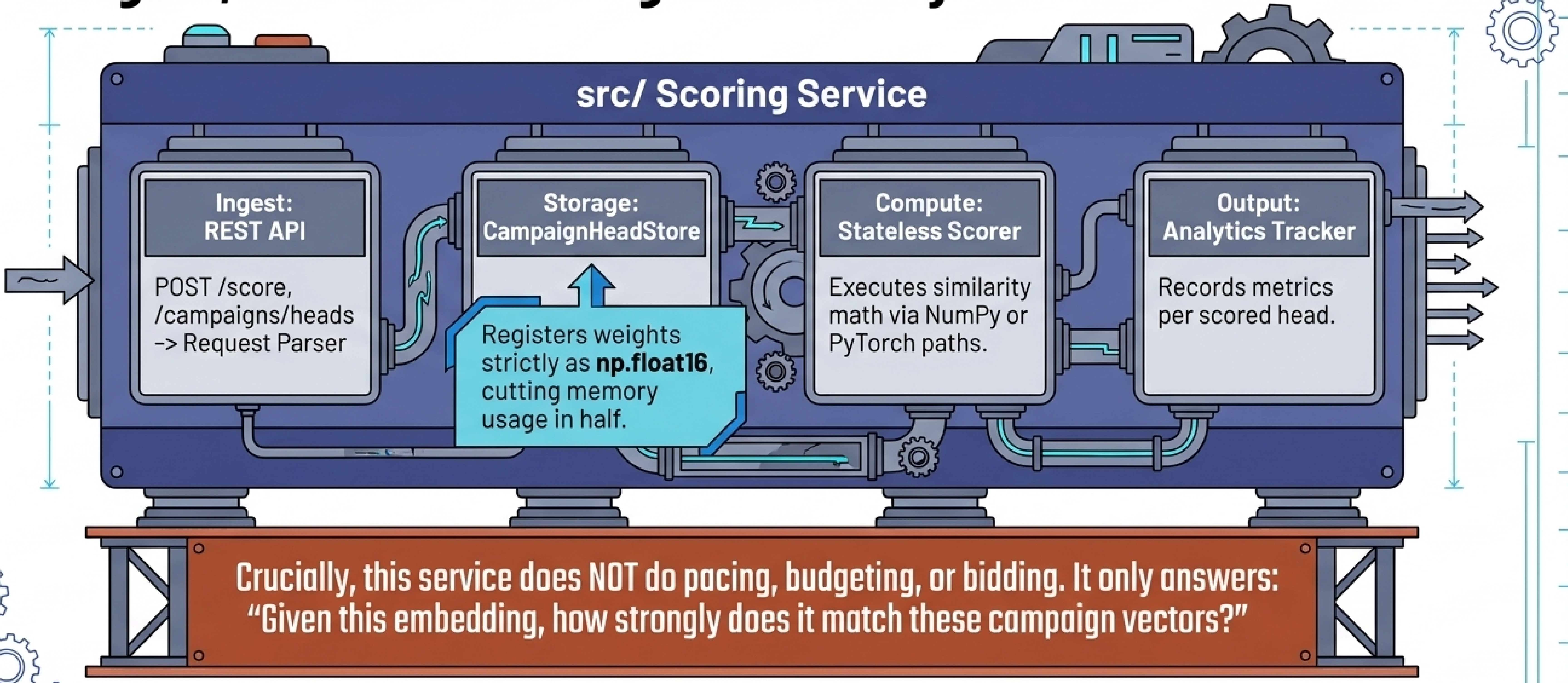
agenticAudienceAdapter.js decodes and maps entries.

Payload additively merged into bid request.

mergeDeep operation: safely coexists with other RTD modules without overwriting the whole user block.

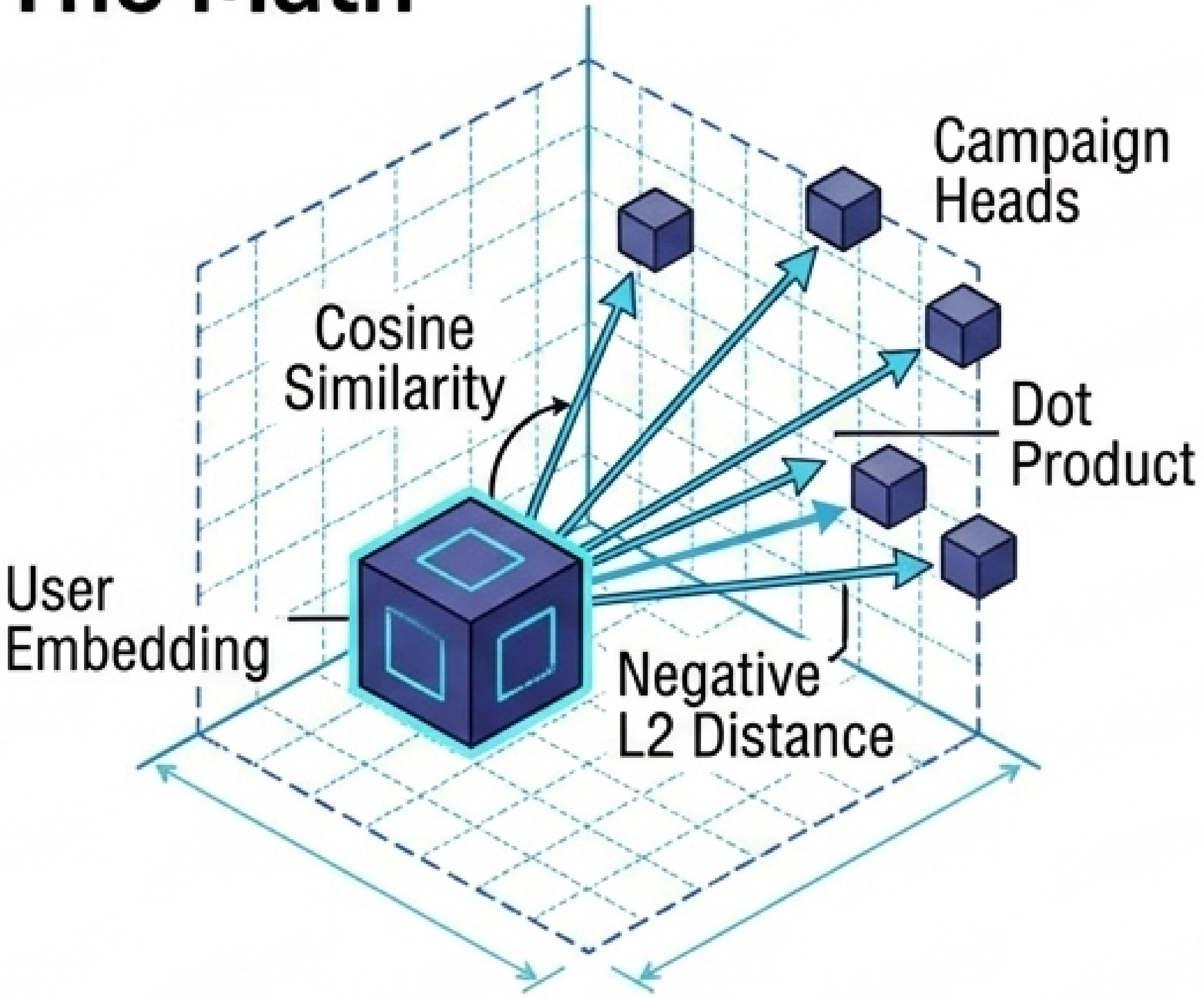
NotebookLM

The FastAPI sidecar is a narrow, highly mature matching engine, not a full bidding decision system.



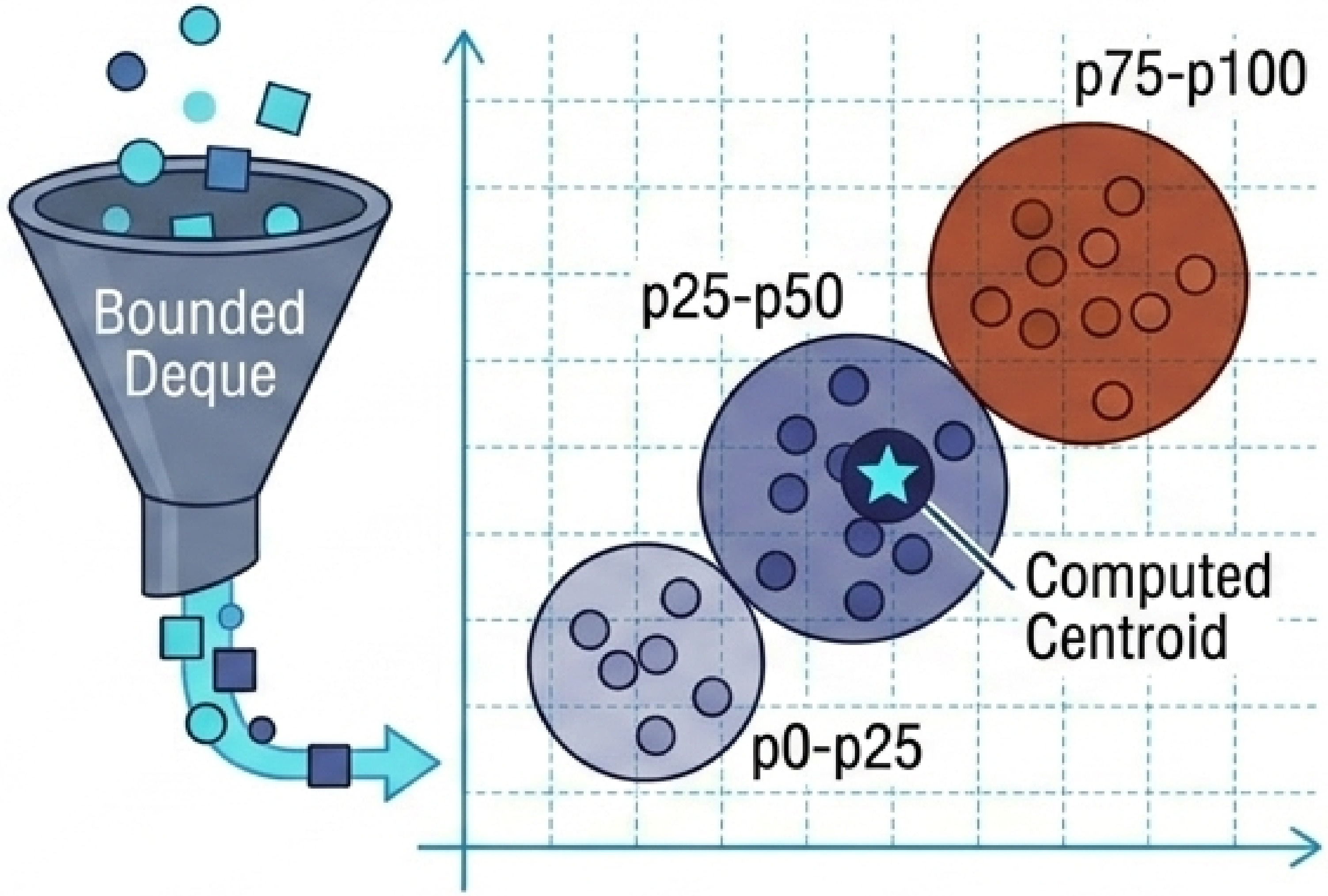
Float16 optimization and bounded PCA analytics make real-time vector math observable.

The Math



Optimized top-k selection via `np.argpartition` (NumPy) or `torch.topk` (PyTorch).

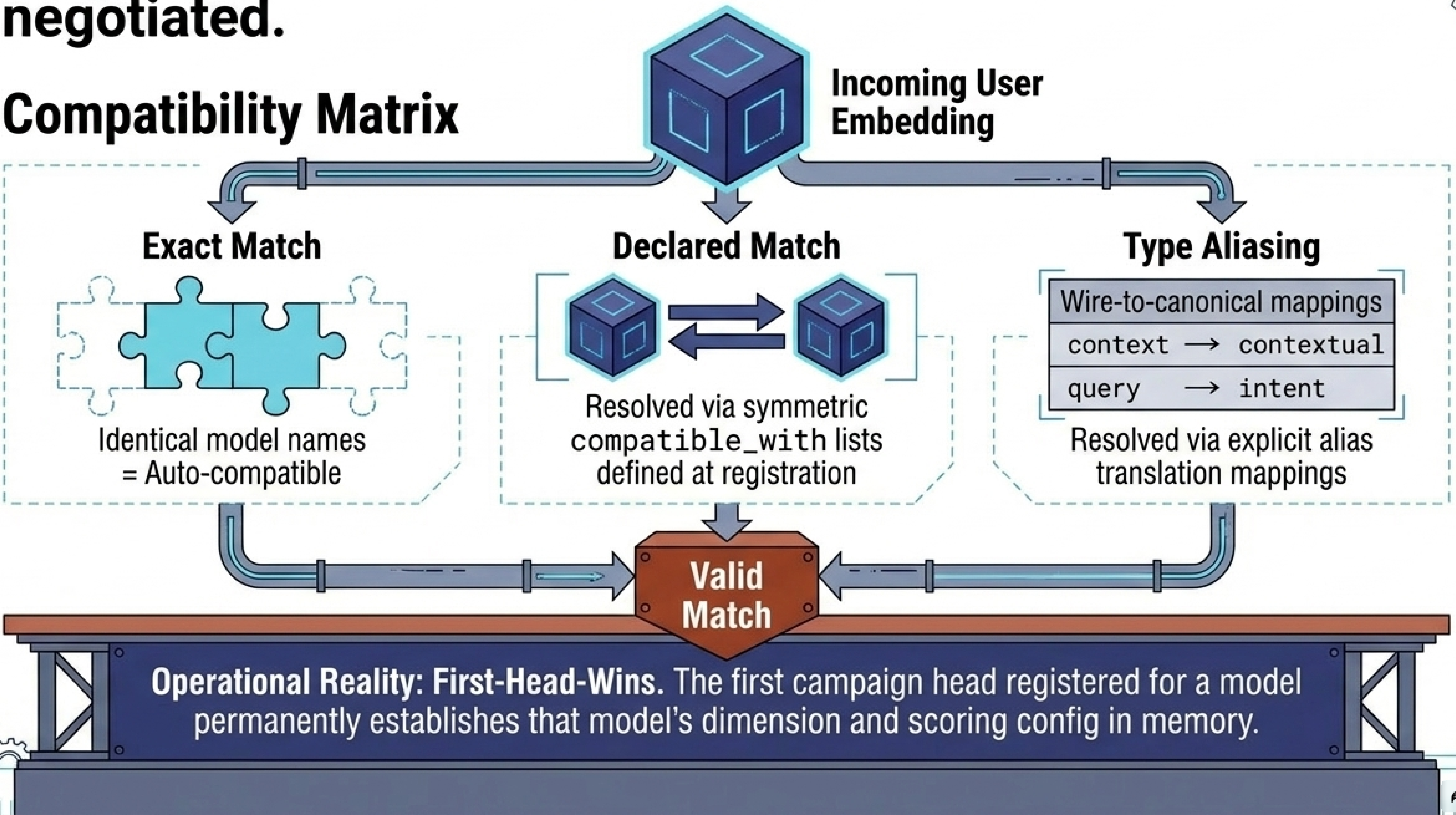
The Analytics



Explains interpretability: visually answers why specific embeddings scored high or low.

Vector compatibility is strictly conditional and explicitly negotiated.

Compatibility Matrix



Distinct design goals create intentional data model mismatches across the repository's layers.

Formal Spec	Prebid Transport	Scoring Service
Rich JSON Envelope	ext.aa	Flattened ext
Spec wire names (e.g., context)	Permissive array/numeric typing	Strict string canonical types
Fully supported	Lacks consent fields	Ignored
embedding_space_id required	Minimal/Transport-oriented	Ignored during runtime math

The **Spec** defines the future contract, Prebid shows one insertion path, and the **Scorer** implements only the narrow math requirements.

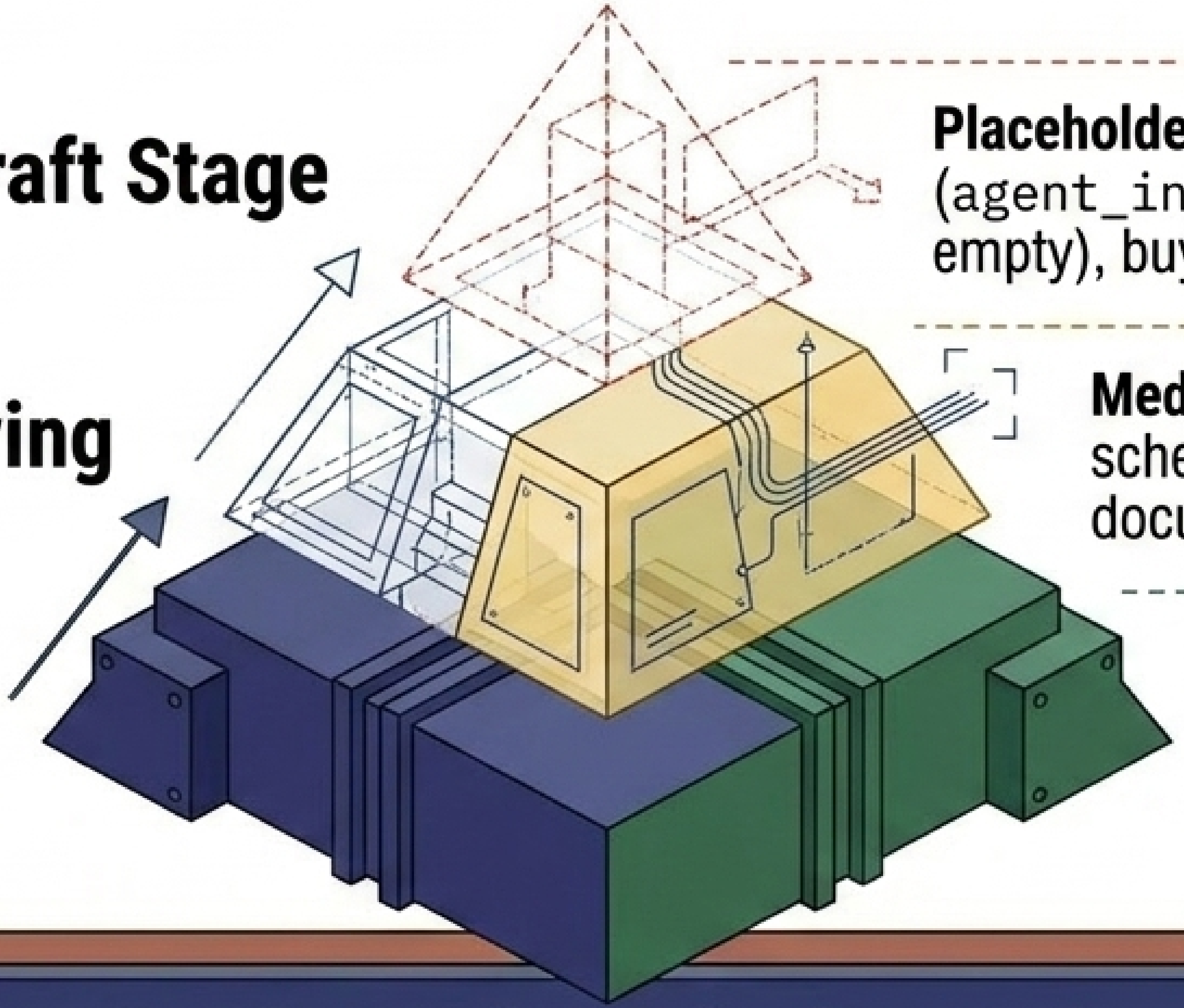
The repository is highly mature in core scoring math, but remains in the draft stage for multi-agent negotiation.

Maturity Map

Draft Stage

Evolving

Highly Mature



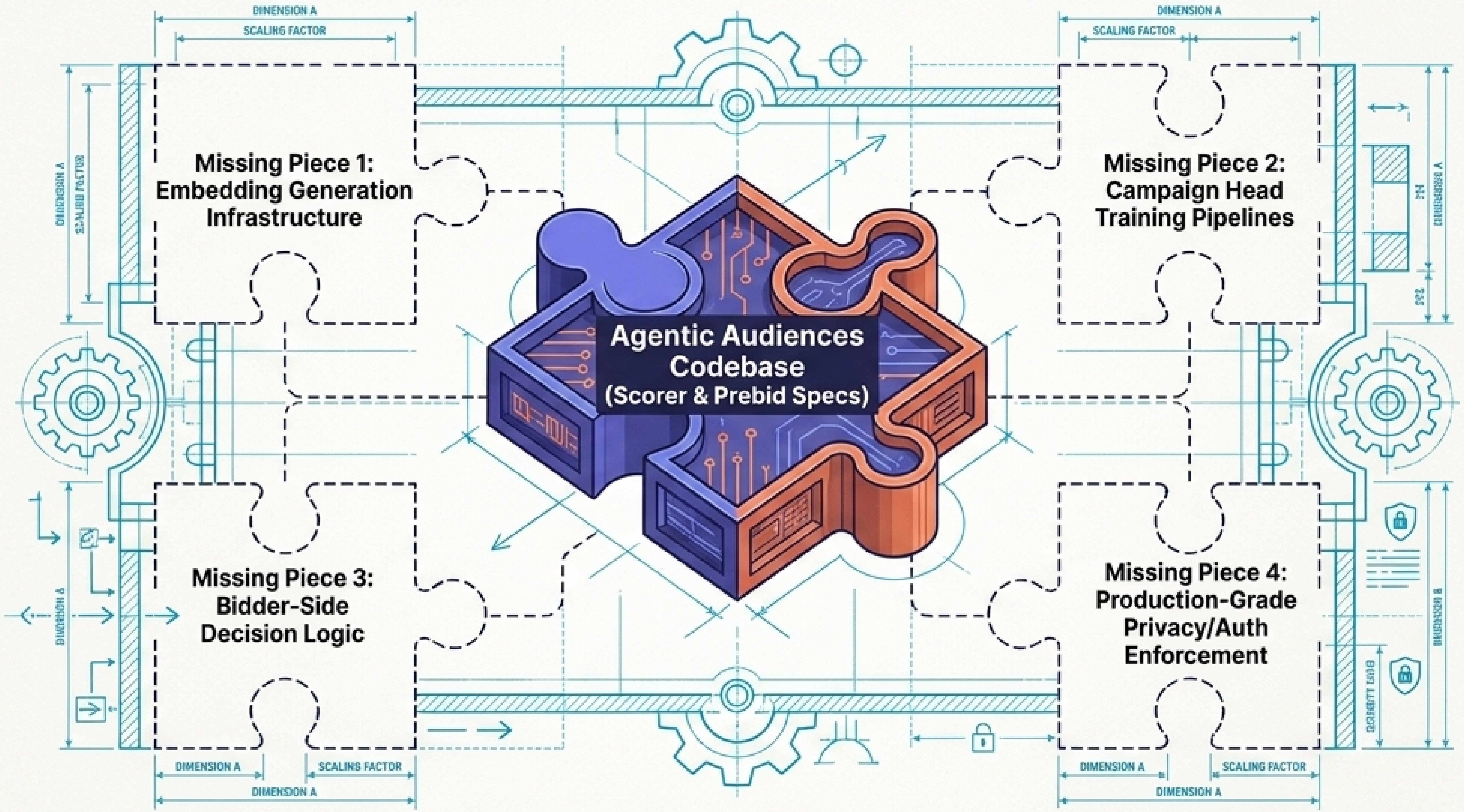
Placeholders: Agent interfaces (`agent_interface.schema.json` is empty), buyer/seller payloads.

Medium Maturity: Embedding schema, architectural documentation, taxonomy.

Production-Ready: FastAPI Scorer, NumPy/Torch fallback logic, test coverage, Prebid payload plumbing.

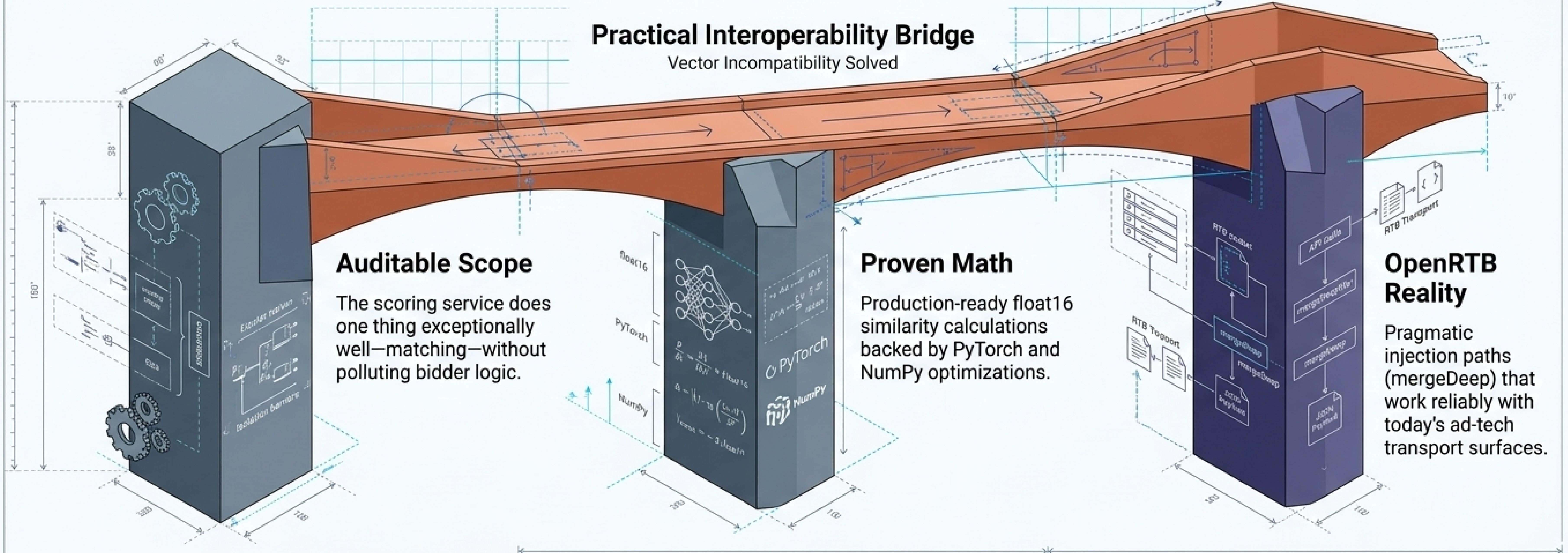
Adopter Reality: The math is tested and proven; the policy and negotiation layers require active community standard-setting.

Deploying this protocol requires robust external infrastructure beyond the open-source code



This repo provides the interoperability standard and DSP sidecar, not a turnkey ad-server.

By treating vector incompatibility seriously and anchoring in OpenRTB, the protocol achieves practical interoperability.



Explore the Code

Read the Taxonomy

Join the Working Group